

ENVARIABELSANALYS DEL 2

SEMINARIUM 5: SAMMANFATTNING

Recap

Rationella funktioner, $\frac{p(x)}{k(x)}$ där $p(x)$
och $k(x)$ är Polynom.

Ex:

$$\int \frac{x}{x^2-1} dx = \int \frac{x}{(x-1)(x+1)} dx$$
$$= \left[\begin{array}{l} \text{Behövs ej se} \\ \text{nämner och} \\ \text{täljare} \end{array} \right] = \int \left(\frac{1}{2} \cdot \frac{2x}{x^2-1} \right) dx$$
$$= \frac{1}{2} \int \frac{2x}{x^2-1} dx = \frac{1}{2} \ln|x^2-1| + C$$

SVAR:

$$F(x) = \frac{1}{2} \ln|x^2-1| + C$$

Ex: $\int \frac{x^2+6x}{(x^2+4)(x-2)} dx = \left[\begin{array}{l} \text{Ansats för} \\ \text{partial bråk} \\ \text{-uppsöcker} \end{array} \right]$

$$\frac{x^2+6x}{(x^2+4)(x-2)} = \frac{Ax+B}{(x^2+4)} + \frac{C}{(x-2)}$$

Hand
-lösning.

[OBS täljare ska vara en
grad mindre än nämnare]

$$C = \frac{16}{8} = 2 \quad \left| = \frac{Ax+B}{(x^2+4)} + \frac{2}{(x-2)} = \frac{(Ax+B)(x-2)+2(x^2+4)}{(x^2+4)(x-2)} \right.$$

$$\Rightarrow x^2+6x = (Ax+B)(x-2) + 2(x^2+4)$$

$$x^2+6x = Ax^2 - 2Ax + Bx - 2B + 2x^2 + 8$$

$$x^2+6x = x^2(A+2) + x(-2A+B) + (-2B+8)$$

Identifiera! $\begin{cases} 1 = A+2 \\ 6 = -2A+B \\ 0 = -2B+8 \end{cases} \longrightarrow$

$$\left[\begin{aligned} & \Leftrightarrow \begin{cases} A = -1 \\ 6 = B + 2 \\ B = 4 \end{cases} \Leftrightarrow \begin{cases} A = -1 \\ B = 4 \\ C = 4 \end{cases} \end{aligned} \right]$$

$$\int \left(\frac{-x+4}{x^2+4} + \frac{2}{x-2} \right) dx$$

$$= \int \left(\frac{-x+4}{x^2+4} \right) dx + \int \frac{2}{x-2} dx$$

$$= \int \frac{-x}{x^2+4} dx + \int \frac{4}{x^2+4} dx + \int \frac{2}{x-2} dx$$

$$= - \int \frac{x}{x^2+4} dx + \int \frac{4}{x^2+4} dx + 2 \int \frac{1}{x-2} dx$$

$$\left[\begin{aligned} & x^2+4 = t \\ & 2x dx = dt \\ & x dx = \frac{1}{2} dt \end{aligned} \right] = \left[\begin{aligned} & \text{eller} \\ & \text{l\"agg till} \\ & 2 \\ & \text{slippen} \end{aligned} \right] \begin{array}{l} \text{men} \\ \text{var} \\ \text{byte} \end{array}$$

$$= - \frac{1}{2} \int \frac{2x}{x^2+4} dx + \int \frac{4}{4} \left(\frac{1}{\left(\frac{t}{2}\right)^2+1} \right) dt + 2 \int \frac{1}{x-2} dx$$

$$\left[\begin{aligned} & \frac{x}{2} = t \\ & dx = 2 dt \end{aligned} \right]$$

$$= - \frac{1}{2} \int \frac{2x}{x^2+4} dx + \int \frac{1}{t^2+1} 2 dt + 2 \int \frac{1}{x-2} dx$$

$$= - \frac{1}{2} \cdot \ln|x^2+4| + 2 \arctan(t) + 2 \ln|x-2| + C$$

$$= - \frac{1}{2} \ln(x^2+4) + 2 \arctan\left(\frac{x}{2}\right) + 2 \ln|x-2| + C$$

Svar:

$f(x) =$

"Hur fungerar partialläggning?"

$$\frac{x^2 + 6x}{(x+4)(x-2)} = \frac{Ax+B}{x+4} + \frac{C}{(x-2)} \quad | \cdot (x-2)$$

Jag vill ha C!

$$\frac{(x^2+6x)(\cancel{x-2})}{(x+4)(\cancel{x-2})} = \frac{(Ax+B)(x-2)}{x+4} + C$$

$$\frac{x^2+6x}{x+4} = \frac{(Ax+B)(x-2)}{(x+4)} + C$$

Sätt nu $x = -4$ då blir C ensamt!

$$\frac{4+16}{4+4} = \frac{(A \cdot -2 + B) \cdot 0}{4+4} + C$$

$$C = \frac{16}{8} = 2$$

MAGGANS Guide!

$$\frac{p(x)}{n(x)}$$

$$\text{grad } p(x) \geq \text{grad } n(x)$$

JA

Polynomdivision.

NEJ

är deriverbar
av nämnaren
täljaren (som
på en konstant)

NEJ

JA

$$\int \frac{p'(x)}{p(x)} dx = \ln|x| + C$$